

Teaching Statement

Teaching Experience

I am currently a Research Assistant Professor at the Hong Kong University of Science and Technology (HKUST), where I am responsible for almost all aspects of the courses I teach, including lecturing, course design, assessment, and examination, as well as supervising undergraduate and graduate research projects. I view teaching as a central part of academic life, and I am confident in teaching across the core computer science curriculum, including programming, software engineering, testing and verification, and AI-assisted development.

My teaching experience spans large undergraduate cohorts and intensive short courses. As an instructor for COMP1021 Introduction to Computer Science (Spring 2025), I taught and managed a class of 110 students from various departments (e.g., School of Engineering, Business School), taking full charge of lecture preparation, live demonstrations, course structure, assessment design, and student support. I also served as a guest instructor for Nankai University's 2025 International Frontier Courses, where I taught Fundamentals of Software Testing to 228 students in an intensive, full-day format. Earlier, I supported several programming courses as a teaching assistant, including COMP1021 Introduction to Computer Science and COMP3021 Java Programming, where I led lab lectures, Q&A sessions, and hands-on programming guidance.

Across these roles, I have developed a teaching style that is structured, interactive, and attentive to diverse learning backgrounds. I design lectures around clear conceptual storylines, integrate visual and real-world examples, and encourage students to think collaboratively through guiding questions. My sessions consistently attracted the highest attendance among parallel sessions, and students frequently commented that the live coding, conceptual clarity, and interactive structure helped them stay engaged and motivated. I have also received strong teaching evaluations (4.22/5), above both the course-wide (4.13) and university-wide (4.19) undergrad averages. The following is excerpted from student feedback:

- *"A very good teaching provided, we have lessons out of interest and glee."*
- *"She taught with examples instead of reading from slideshow, so it was more engaging and helpful."*
- *"10/10 teaching, despite being fairly new to teaching."*
- *"good teaching! excellent organized and customized material get me rid of some messy party of the textbook love!!! great teaching and I will not feel bored in class."*

Interviewing and Admission Experience

I also have direct experience interviewing prospective undergraduate students. At HKUST, I served as an interviewer for the JUPAS (Joint University Programmes Admissions System) admissions cycle in 2025, where I was responsible for conducting multiple in-person interview sessions. Each interviewer conducted around four 30-minute sessions, assessing five students per session, as part of an event with over 900 prospective undergrads. I evaluated applicants using structured scoring criteria and detailed student profiles, and contributed to discussions on academic potential, communication skills, and suitability for the School of Engineering. This experience has equipped me with the confidence and judgement required for the undergraduate admissions, including structured interviews, academic assessment, and identifying students with strong potential for tutorial-style learning.

In addition, as a Research Assistant Professor, I have acted as a co-examiner for PhD interviews within our research group, assessing applicants' research preparedness, intellectual fit, and long-term potential.

Supervision of Undergraduate and MSc Projects

I have substantial experience supervising student research and capstone work. I have served multiple times as a supervisor and examiner for undergraduate Final-Year Projects, guiding students through problem formulation, research design, and technical implementation, as well as assessing their written reports and oral defences. I am comfortable mentoring both individual and group projects, ensuring that students develop not only technical competence but also the ability to communicate rigorously through presentations and academic writing. I am confident in supervising BA, MEng, and MSc projects across software engineering, testing and verification, and AI-assisted development.

Industrial Experience and Teaching for Professional Learners

My teaching is also informed by significant industry experience. During my PhD, I completed an eighteen-month research internship in a software development company, where I worked on real-world software engineering and AI projects. I currently serve as an industry consultant, providing monthly technical guidance, training sessions, and code-review support to professional software engineers. This background enables me to design and deliver modules that combine rigorous theoretical foundations with practical industry relevance, making me particularly well-prepared to

teach part-time MSc students and professionals in the IT sector. I am comfortable taking full responsibility for setting up and running a module, including designing assignments and projects that connect academic concepts to realistic engineering scenarios.

Assessment and Examining

I have experience with both formative and summative assessment. As part of my teaching assistant roles, I contributed to the preparation and vetting of exam papers, ensuring alignment with learning outcomes and appropriate discrimination across ability levels. As an instructor, I have also spent substantial time explaining past exam questions during lectures and tutorials, which has given me a strong sense of how to design assessments that are fair, challenging, and pedagogically effective. I am prepared to take responsibility for examining, marking, and quality assurance for undergraduate and graduate modules.

Supervision of Postgraduates

I have substantial experience mentoring students at the master's and doctoral levels, developed through my roles as a senior PhD student, postdoctoral researcher, and now Research Assistant Professor. My supervision approach combines clear intellectual guidance with structured milestones, helping students formulate meaningful research questions, design rigorous methodologies, and communicate their contributions through publications and dissertations.

At the graduate level, I have co-supervised five PhD students and four MSc students from institutions including HKUST, Nanjing University, the Chinese Academy of Sciences, Huazhong University of Science and Technology, and Zhejiang University. My responsibilities have included shaping research direction, providing technical and writing feedback, conducting regular research meetings, and preparing students for conference submissions. These collaborations have produced six published papers to date, with several additional manuscripts currently under review.

Overall, my teaching and mentoring experience has equipped me to support students with diverse backgrounds and levels of preparation, and I am fully prepared to supervise undergraduate, MEng, and MSc projects independently.